CLAIMS

What is claimed is:

1. A data transmission system, comprising:

    - a transmitter unit including a two-dimensional array of individually addressable electromagnetic radiation emitters;
    - an encoder circuit configured to receive digital data and to generate a sequence of frames by mapping said digital data to the emission characteristics and spatial positions of the emitters in the array;
    - a receiver unit including an array of electromagnetic radiation sensors configured to capture the sequence of frames emitted by the transmitter unit;
    - a decoder circuit configured to generate a replica of the digital data by interpreting the emission characteristics and spatial positions of the emitters in the captured frames; and
    - a physical housing assembly configured to optically couple the transmitter unit and the receiver unit while shielding them from ambient radiation.

2. The system of claim 1, wherein the emission characteristics include at least one of wavelength or intensity.

3. The system of claim 1, wherein the electromagnetic radiation emitters and sensors operate in the visible light spectrum.

4. The system of claim 3, wherein the transmitter unit is a micro-display and the receiver unit is an image sensor, and wherein the emission characteristics are the color and brightness of pixels.

5. The system of claim 1, wherein the electromagnetic radiation emitters and sensors operate in a non-visible spectrum, such as infrared or ultraviolet.

6. The system of claim 1, wherein the data transmission bandwidth of the system is a direct function of the number of emitters in the array.

7. The system of claim 1, further comprising a communication protocol wherein the receiver unit is configured to query the transmitter unit to determine a set of emission capabilities and to calibrate the decoder circuit based on said capabilities.

8. The system of claim 1, wherein the encoder circuit is configured to form a "channel element" by assigning the same emission characteristics to a predefined block of adjacent emitters in the array.

9. A method for transmitting data, comprising the steps of:

    - receiving, at a transmitter, a stream of digital data;
    - encoding said data into a sequence of frames, wherein each frame comprises a plurality of emission channels, and wherein each channel is assigned an emission characteristic corresponding to a portion of the data;
    - emitting said sequence of frames from a two-dimensional array of emitters;
    - capturing said sequence of frames with an array of sensors optically coupled to the array of emitters;
    - decoding the captured frames by reading the emission characteristic and position of each channel to reconstruct the stream of digital data; and
    - outputting, from a receiver, the reconstructed stream of digital data.

10. The method of claim 9, wherein the emission characteristic is selected from the group consisting of: wavelength, intensity, color, and brightness.

11. The method of claim 9, further comprising the preliminary step of:

- transmitting, from the transmitter to the receiver, a set of parameters describing the capabilities of the emitter array; and
- calibrating, at the receiver, a decoding map based on said parameters.